

SIT720 — 11.1 HD TASK

Machine Learning Mini Research

Option 3: An efficient stacking-based ensemble technique for early heart attack prediction

Technical Research Report

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Video presentation: [Add final video link here]

Code/data archive: [Add final GitHub, OneDrive or Dropbox link here]

Abstract

This study reproduces the machine-learning experiments reported by Bhagat, Sharma and Agarwal [1] and then redesigns the evaluation protocol to address a data-integrity weakness identified in the supplied dataset. The dataset contains 1,025 rows and 14 columns, but 723 rows are exact duplicates, leaving 302 unique rows. Under the seed-42 paper-style split, 202 test observations have an identical predictor vector in the training data. The paper-style stacking reproduction reaches 1.0000 accuracy, while the matched ten-seed paper-style protocol averages 0.9971 ± 0.0062 compared with 0.8492 ± 0.0603 for the leakage-aware stacking protocol. The mean accuracy gap is 14.8 percentage points, with paper-style accuracy higher in all 10 of 10 nominal seed cases. An unpaired Mann–Whitney U test gives $p < 0.001$; the Wilcoxon $p=0.0020$ is retained only as a sensitivity calculation because the underlying test sets differ. A Leakage-Aware Stability-Selected Ensemble (LASSE) is then evaluated. LASSE does not significantly outperform the leakage-aware stacking baseline, so the contribution is framed as improved experimental validity rather than an unsupported classifier improvement.

1. Introduction and research problem

Heart-disease prediction is a binary classification problem in which the target indicates the presence or absence of disease. The selected paper investigates this problem with 13 clinical predictors and six conventional classifiers, followed by a 5-fold stacking ensemble [1]. The paper reports 98.53% stacking accuracy and presents the ensemble as its main contribution. The reproduction study first asks whether the published results can be obtained from the supplied dataset using a clearly documented interpretation of the paper. It then asks whether those results remain credible when duplicate observations and model-selection risks are controlled.

The central methodological issue is the independence of the evaluation cases. The audit found 723 exact duplicate rows. With a random row-level split, identical predictor vectors can occur in both training and test sets. In the seed-42 split, 202 test observations have an identical predictor vector in training. Kapoor and Narayanan identify leakage as a source of overoptimistic machine-learning results and emphasise reproducible evaluation practices [5]. This provides a concrete reason to separate the faithful reproduction of the published protocol from a stricter leakage-aware experiment.

The research gap addressed here is therefore methodological. The selected paper does not document duplicate control, gives incomplete implementation details, and reports internally inconsistent metric values. The proposed study makes the data unit, preprocessing, model selection, feature-selection stability and repeated evaluation explicit. The objective is not to manufacture a higher score, but to determine how much the apparent performance depends on the evaluation protocol and whether a redesigned pipeline provides a more defensible estimate of performance.

2. Selected paper and state-of-the-art context

Bhagat, Sharma and Agarwal use a 1,025-observation dataset with 14 attributes: 13 predictors and a binary target [1]. The paper evaluates Logistic Regression, Decision Tree, Random Forest, XGBoost, Naive Bayes and K-Nearest Neighbors (KNN), followed by a 5-fold stacking ensemble. The paper reports 92.68% for both Decision Tree and Random Forest, 90.73% for XGBoost and 98.53% for stacking in its main results. Its related-work section also reports results from HRFLM, CHI+PCA, Lasso/Relief with Random Forest, FCMIM-SVM and other ensemble approaches [1].

Those published studies use different datasets, feature-selection strategies and validation protocols, so their accuracies should not be interpreted as a single directly comparable ranking. The selected paper itself reports two conflicting accuracy statements: XGBoost is described once as 93.17% and elsewhere as 90.73%, while stacking is described once as 96.58% and elsewhere as 98.53% [1]. This report uses Table 11 of [1] as the principal numerical benchmark because it is repeated in the main results and later comparison tables.

Table 1. Accuracy results reported for related approaches in the selected paper.

Related study / approach	Reported accuracy
Mohan et al.	88.47%
Ramesh et al.	88.35%
Alotaibi	93.19%
Li et al. (FCMIM-SVM)	92.37%
Ali et al.	92.22%
Latha et al.	85.48%

These published results provide context rather than a directly comparable benchmark because the studies use different datasets, feature sets and validation protocols. The leakage-aware estimate of about 85% in this study falls within the broad 85–93% range reported by these related studies, whereas the paper’s 98.53% stacking result is substantially higher. The difference is therefore consistent with the protocol concerns identified here, but it does not by itself establish that one study is intrinsically more accurate than another.

Three methodological strands motivate Part 2. Leakage can make empirical performance appear stronger than the underlying generalisation ability [5]. Model selection can also introduce selection bias when a finite sample is repeatedly used to choose among models and settings [6]. Stability selection uses resampling to assess whether variables are selected consistently rather than relying on a single fitted subset [7]. The proposed design combines these ideas with a duplicate-aware data split and fold-safe preprocessing.

3. Dataset and feature set

The paper states that the broader source contains 76 characteristics, but the experimental analysis uses 14 attributes, including the target [1]. The 13 predictors used here are reproduced below. The ranges are those reported in the paper; they are also useful for identifying the anomalous $ca=4$ and $thal=0$ codes observed in the supplied file. Predictor types follow the implementation used in Part 2: five continuous variables are scaled numerically, while eight nominal/binary variables are one-hot encoded.

Predictor	Description	Type	Paper range
age	Age in years	Numeric	29–77
sex	Sex/gender coding	Categorical	0, 1
cp	Chest-pain type	Categorical	0–3
trestbps	Resting blood pressure (mmHg)	Numeric	94–200
chol	Serum cholesterol (mg/dL)	Numeric	126–564
fbs	Fasting blood sugar >120 mg/dL	Categorical	0, 1
restecg	Resting ECG result	Categorical	0–2
thalach	Maximum heart rate achieved	Numeric	71–202
exang	Exercise-induced angina	Categorical	0, 1
oldpeak	ST depression induced by exercise	Numeric	0–6.2
slope	Slope of peak exercise ST segment	Categorical	0–2
ca	Number of major vessels coloured by fluoroscopy	Categorical	0–3
thal	Thalassemia coding	Categorical	1–3

Table 2. Predictor set used in the reproduction and proposed experiments; ranges are adapted from Table 9 of [1].

The target is binary: 0 denotes no disease and 1 denotes disease [1]. The supplied file contains no missing cells, but it contains 18 $ca=4$ records and 7 $thal=0$ records, which fall outside the ranges reported by the paper. These values are retained for the faithful Part 1 reproduction and converted to missing values before Part 2 imputation.

3.1 Data audit and duplicate leakage

Audit measure	Value
Rows	1025
Columns	14
Predictors	13
Missing cells	0
Exact duplicate rows	723
Unique rows	302
Class 0	499
Class 1	526
$ca=4$ records	18
$thal=0$ records	7

Table 3. Initial audit of the supplied heart.csv file.

The duplicate count is not merely a data-cleaning detail. A separate diagnostic compares predictor vectors across the seed-42 train/test split. It finds 202 test observations whose predictor vector also occurs in training, and no duplicated predictor group has conflicting target labels. This means the test set is not independent at the level of unique predictor cases. The exact duplication provides a plausible mechanism for memorisation by flexible tree-based models.

4. Part 1 — faithful reproduction

4.1 Experimental protocol

The paper does not specify the exact random seed, a complete numerical hyperparameter table, the exact categorical encoder, or a duplicate-handling rule. The reproduction therefore makes these assumptions explicit: a stratified 80:20 split with seed 42; median imputation and standardisation inside a pipeline; the published 13 predictors; library-default classifier settings unless a parameter is explicitly required by the implementation; and a 5-fold stacking classifier with logistic regression as the meta-learner using probability outputs. The six individual models are Logistic Regression, Decision Tree, Random Forest, XGBoost, Gaussian Naive Bayes and KNN.

Underspecified element	Choice	Rationale
Train/test split	Stratified 80:20, seed 42	Reproducible binary-classification split preserving class proportions.
Hyperparameters	Library defaults unless explicitly set	The paper does not provide a complete numerical parameter table.
Preprocessing	Median imputation + standardisation inside pipelines	Matches the paper description while keeping transformations within training data.
Categorical encoding	Integer-coded predictors retained in Part 1	The paper says categories are encoded but does not identify the encoder; retaining the 13-column representation is the closest reproducible interpretation.
Stacking	5 folds, probability outputs, logistic-regression meta-learner	5 folds are stated by the paper; logistic regression is one of the meta-model examples described by the authors.
Duplicate handling	None in Part 1	Faithful reproduction must preserve the original row-level protocol.
Part 2 split	Exact duplicates removed before splitting	Creates an evaluation set based on unique predictor cases.
Part 2 preprocessing	Invalid codes → missing; one-hot categoricals; scaling of continuous features	Addresses data validity and nominal-variable representation.
Primary tuning	30 random configurations/model, 5-fold CV	Broader search while keeping the test set untouched.
Repeated LASSE tuning	15 random configurations/model, 3-fold CV within each seed training set	Prevents seed-42 tuning choices from entering later test sets.
Stability selector	5 resampling repetitions	Stability-inspired rather than a full classical stability-selection implementation.
Repeated baseline	Leakage-aware stacking uses defaults; LASSE is tuned	Asymmetric comparison is declared as a limitation rather than hidden.

Table 4. Explicit experimental assumptions and design choices.

4.2 Machine-learning methods

Method	Role and implementation
Logistic Regression	Linear probabilistic classifier for binary outcomes; used with standardised inputs and default regularisation settings in Part 1.
Decision Tree	Recursive tree classifier that partitions observations using feature-based rules; default scikit-learn settings are used.
Random Forest	Bagged collection of decision trees whose predictions are aggregated; 100 trees are used in the submitted implementation.
XGBoost	Gradient-boosted decision-tree classifier that adds trees sequentially to reduce the loss; the implementation uses XGBoost's binary objective and log-loss evaluation metric [3].
Naive Bayes	Probabilistic classifier based on conditional independence assumptions between predictors.
KNN	Instance-based classifier that assigns a class using the nearest training observations.
5-fold stacking	Base-model probability predictions are generated through cross-validation and supplied to a logistic-regression meta-learner. This follows the stacked-generalization idea introduced by Wolpert [2].

Table 5. Machine-learning methods used in the paper-style reproduction.

4.3 Evaluation metrics

Metric	Formula / definition	Interpretation
Accuracy	$(TP+TN)/(TP+TN+FP+FN)$	Overall proportion classified correctly.
Precision	$TP/(TP+FP)$	Proportion of predicted positives that are true positives.
Recall / Sensitivity	$TP/(TP+FN)$	Proportion of actual positive cases detected.
F1	$2TP/(2TP+FP+FN)$	Harmonic mean of precision and recall.
AUC	Area under ROC curve	Measures ranking discrimination across classification thresholds.
Specificity	$TN/(TN+FP)$	Proportion of actual negative cases correctly identified.
MCC	$(TP \cdot TN - FP \cdot FN) / \sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}$	Correlation-style summary using all four confusion-matrix cells.

Table 6. Evaluation metrics used in the reproduction and extended analysis.

4.4 Reproduced results

Model	Accuracy	Precision	Recall	F1	AUC	Spec.	MCC
Logistic Regression	0.8098	0.7619	0.9143	0.8312	0.9298	0.7000	0.6309
Decision Tree	0.9854	1.0000	0.9714	0.9855	0.9857	1.0000	0.9712
Random Forest	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
XGBoost	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
Naive Bayes	0.8293	0.8070	0.8762	0.8402	0.9043	0.7800	0.6602
KNN	0.8634	0.8738	0.8571	0.8654	0.9629	0.8700	0.7269
Stacking	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Table 7. Paper-style reproduction results on the seed-42 80:20 split.

Random Forest, XGBoost and stacking each reach 1.0000 accuracy on the supplied file. Decision Tree reaches 0.9854, while Logistic Regression and Naive Bayes remain below the published values. The contrast is important: flexible tree-based methods can exploit repeated predictor vectors much more directly than the more constrained probabilistic and linear models. This pattern supports the duplicate-leakage explanation rather than a generic claim that every model is inflated.

Figure 1. ROC curves for the seven paper-style reproduced models; the Random Forest, XGBoost and stacking curves overlap at $AUC = 1.000$ and therefore are visually superimposed.

Figure 2. Confusion matrices for the six reproduced individual classifiers.

4.5 Published versus reproduced results

Model	Pub. Acc.	Rep. Acc.	Δ Acc.	Pub. F1	Rep. F1	Pub. AUC	Rep. AUC
Logistic Regression	0.8439	0.8098	-0.0341	0.8620	0.8312	0.9204	0.9298
Naive Bayes	0.8439	0.8293	-0.0146	0.8608	0.8402	0.9185	0.9043
KNN	0.8585	0.8634	+0.0049	0.8687	0.8654	0.9304	0.9629
Decision Tree	0.9268	0.9854	+0.0586	0.9289	0.9855	0.9702	0.9857
Random Forest	0.9268	1.0000	+0.0732	0.9333	1.0000	0.9734	1.0000
XGBoost	0.9073	1.0000	+0.0927	0.9132	1.0000	0.9830	1.0000
Stacking	0.9853	1.0000	+0.0147	0.9861	1.0000	0.9880	1.0000

Table 8. Side-by-side published and reproduced performance; Δ Accuracy = reproduced – published.

The reproduced results differ materially from Table 11 of [1]. XGBoost has the largest accuracy increase (+0.0927), followed by Random Forest (+0.0732) and Decision Tree (+0.0586). Logistic Regression and Naive Bayes are lower than the paper, while KNN is close to the published value. Stacking is 0.0147 above the published 0.9853. The direction is therefore model-dependent, but the exact 1.0000 results for the tree-based models and stacking are particularly consistent with duplicated observations crossing the split.

The paper also contains two explicit reporting conflicts. XGBoost is stated as 93.17% in one passage but 90.73% in the main results, and stacking is stated as 96.58% in one passage but 98.53% in Table 11 of [1], Table 13 of [1] and the conclusion [1]. These values are retained as reporting inconsistencies rather than silently reconciled.

4.6 Internal consistency of the proposed-model metrics

Figure 5 of the paper reports $TN=95$, $FP=0$, $FN=3$ and $TP=107$. Recomputing the metrics from those counts gives the values below. The first four measures agree with the rounded published values, but specificity and MCC do not.

Metric	Recomputed from Fig. 5	Published	Difference
Accuracy	0.9854	0.9853	+0.0001
Precision	1.0000	1.0000	+0.0000
Recall/Sensitivity	0.9727	0.9727	+0.0000
F1	0.9862	0.9861	+0.0001
Specificity	1.0000	0.9861	+0.0139
MCC	0.9711	1.0000	-0.0289

Table 9. Independent consistency check of the paper's reported stacking metrics.

Specificity computed from Fig. 5 is 1.0000 because there are no false positives: $95/(95+0)=1$. The published specificity value of 0.9861 therefore cannot be obtained from the displayed confusion matrix. MCC is 0.9711 from the same four cells, not 1.0000. Table 11 of [1] also repeats accuracy in the sensitivity column, F1 in the specificity column and precision in the MCC column for the proposed model. This is a concrete internal reporting inconsistency and reduces confidence in treating the table as a self-contained ground truth.

5. Part 2 — Leakage-Aware Stability-Selected Ensemble (LASSE)

5.1 Motivation and methodological contribution

LASSE means Leakage-Aware Stability-Selected Ensemble. The contribution is deliberately broader than changing a classifier: the evaluation level, invalid-code handling, categorical representation, feature-selection procedure, tuning protocol and repeated assessment are redesigned together. The method first removes exact duplicate rows, then converts $ca=4$ and $thal=0$ to missing values, preprocesses variables according to type, applies a resampling-based L1 feature-selection step and combines four tuned learners with soft voting.

The design follows leakage-control and model-selection principles [5], [6] and uses selection frequency as a stability signal [7]. The selector is described as stability-inspired rather than as a full classical stability-selection implementation because it uses five repetitions of stratified subsampling rather than the much larger resampling regimes normally associated with the original stability-selection formulation [7].

Two expectations are stated before the results are examined. H1: the row-level evaluation protocol overstates performance compared with the leakage-aware protocol. H2: LASSE improves performance over leakage-aware stacking.

5.2 Data preparation and preprocessing

Exact duplicate rows reduce the dataset from 1,025 to 302 unique rows before the Part 2 split. The 18 $ca=4$ and 7 $thal=0$ observations are converted to missing values. Numerical predictors are median-imputed and standardised. Nominal predictors are imputed with the most frequent category and one-hot encoded. All transformations are inside scikit-learn pipelines, so the test set is not used to fit preprocessing parameters [4].

5.3 Stability selection and ensemble design

After preprocessing, the stability selector repeatedly fits an L1-regularised logistic model on stratified subsets of the training data. A transformed feature is retained when its selection frequency reaches the tuned threshold. The final soft-voting ensemble combines Logistic Regression, RBF-SVM, Random Forest and KNN. Each component is tuned using F1 as the cross-validation objective. The primary search samples 30 configurations per model family using 5-fold stratified cross-validation.

The repeated evaluation uses a stricter protocol: each of the ten seeds creates a new training/test split, and LASSE is retuned from scratch inside that training partition. This prevents the seed-42 hyperparameter choices from being reused on later test sets, directly addressing model-selection bias [6].

5.4 Tuning results

Model	Best CV F1	Configs	Selected parameters
Logistic Regression	0.8802	30	{'select__frequency_threshold': 0.5, 'select__C': 1.0, 'model__C': 1}
RBF-SVM	0.8866	30	{'select__frequency_threshold': 0.6, 'select__C': 1.0, 'model__gamma': 'auto', 'model__C': 1}
Random Forest	0.8697	30	{'select__frequency_threshold': 0.5, 'select__C': 1.0, 'model__n_estimators': 200, 'model__max_features': 'sqrt', 'model__max_depth': 5}
KNN	0.8626	30	{'select__frequency_threshold': 0.7, 'select__C': 1.0, 'model__weights': 'uniform', 'model__n_neighbors': 9}

Table 10. Primary tuning summary for the four LASSE component models.

5.5 Primary clean-data comparison and fair ablation

Model	Accuracy	Precision	Recall	F1	AUC	Spec.	MCC
Leakage-aware stacking	0.8197	0.8438	0.8182	0.8308	0.8896	0.8214	0.6382
Proposed LASSE	0.8197	0.8235	0.8485	0.8358	0.8907	0.7857	0.6363
Ablation: same voting without stability selection	0.8197	0.8438	0.8182	0.8308	0.8939	0.8214	0.6382

Table 11. Primary leakage-aware holdout comparison and fair no-selection ablation.

Table 12. Direct comparison of LASSE with the reproduced baseline and the published stacking result.

Model / reference	Accuracy	Precision	Recall	F1	AUC
Published stacking [1]	0.9853	1.0000	0.9727	0.9861	0.9880
Reproduced paper-style stacking (seed 42)	1.0000	1.0000	1.0000	1.0000	1.0000
Leakage-aware stacking (clean holdout)	0.8197	0.8438	0.8182	0.8308	0.8896
LASSE (clean holdout)	0.8197	0.8235	0.8485	0.8358	0.8907

The first two rows use the duplicated row-level protocol used for the paper-style reproduction, whereas the bottom two rows use the leakage-aware protocol. The difference between these groups therefore

reflects the evaluation protocol as well as the model, while LASSE and leakage-aware stacking form the like-for-like model comparison.

The fair ablation changes only one component: it removes the stability-selection step while retaining the same four-model soft-voting architecture and retuning the component models on the same training partition. The ablation and leakage-aware stacking baseline both reach 0.8197 accuracy, so the selector does not add accuracy on this holdout. LASSE nevertheless changes the error profile: recall increases from 0.8182 to 0.8485, while specificity decreases from 0.8214 to 0.7857. For a screening-oriented application, that trade-off can be relevant because higher recall means fewer positive cases are missed, although the corresponding increase in false positives must also be considered. The result is therefore a trade-off, not evidence of overall superiority.

Figure 3. ROC comparison under the leakage-aware protocol.

Figure 4. Confusion matrices for leakage-aware stacking and LASSE.

5.6 Repeated evaluation and matched protocol comparison

The clean-data model comparison is repeated over ten stratified 80:20 splits. LASSE is retuned within each training split. The leakage-aware stacking baseline uses its default component settings, so this comparison is asymmetric; that limitation is declared rather than hidden. The repeated model comparison gives 0.8492 ± 0.0603 accuracy for leakage-aware stacking and 0.8443 ± 0.0537 for LASSE. The difference is not statistically supported: Wilcoxon $p=0.4062$ and Nadeau–Bengio corrected $p=0.8155$ [8].

Metric	Paper-style mean $\pm$ SD	Leakage-aware mean $\pm$ SD	Mean gap	Wilcoxon p (sensitivity)	Mann–Whitney p
Accuracy	0.9971 ± 0.0062	0.8492 ± 0.0603	0.1479	0.0020	< 0.001
F1	0.9971 ± 0.0060	0.8633 ± 0.0514	0.1339	0.0020	< 0.001
AUC	0.9999 ± 0.0003	0.9109 ± 0.0464	0.0890	0.0020	< 0.001

Table 13. Matched ten-seed comparison of the paper-style and leakage-aware protocols.

The matched comparison uses the same ten nominal random seeds for both protocols, but the protocols operate on different row-level datasets after duplicate removal. The ten paper-style accuracies exceed the leakage-aware accuracies in all 10 of 10 cases. The two-sided Wilcoxon calculation gives $p=0.0020$, the smallest attainable two-sided value with ten non-zero pairs; because the underlying test observations are not shared, this is reported only as a sensitivity calculation rather than as valid paired evidence. An unpaired two-sided Mann–Whitney U test gives $p < 0.001$. The mean accuracy gap is 0.1479, or 14.8 percentage points. Together, these results provide consistent evidence that the row-level protocol produces a materially more optimistic estimate on this duplicated dataset, while the inferential limitation of the nominal pairing is made explicit.

5.7 Feature-stability evidence

Transformed feature	Selection frequency	Selected
num age	1.0	Yes
num trestbps	1.0	Yes
num thalach	1.0	Yes
num oldpeak	1.0	Yes
cat cp 2.0	1.0	Yes
cat sex 0.0	1.0	Yes
cat sex 1.0	1.0	Yes
cat cp 0.0	1.0	Yes
cat slope 2.0	1.0	Yes
cat ca 0.0	1.0	Yes
cat ca 2.0	1.0	Yes
cat exang 1.0	1.0	Yes
cat thal 3.0	1.0	Yes
num chol	0.8	Yes
cat slope 1.0	0.6	Yes
cat restecg 1.0	0.6	Yes

Table 14. Highest-frequency transformed features in the final LASSE fit.

The selector retains age, resting blood pressure, maximum heart rate, oldpeak, several chest-pain levels, sex, slope, ca and exercise-induced angina with high frequency. Exang and thalach (maximum heart rate) are both retained, which is notable because the paper’s feature-importance discussion places exercise-induced angina and maximum heart rate among its strongest predictors [1]. The output is reported using actual transformed feature names, making the result auditable. Binary one-hot encoding retains both levels for some variables, such as sex; this is a redundancy that could be removed in a future version with `drop="if_binary"`.

6. Critical discussion

6.0 Revisiting the expectations

The two expectations can now be revisited. H1 is supported: across all ten nominal seed cases, paper-style accuracy was higher than leakage-aware accuracy, with the Mann–Whitney U comparison giving $p < 0.001$. H2 is not supported: LASSE did not show a statistically significant improvement over leakage-aware stacking; the repeated comparison gives Wilcoxon $p=0.41$ and Nadeau–Bengio corrected $p=0.82$.

6.1 What the reproduction establishes

The reproduction establishes that the published performance level is attainable, and even exceeded, under a row-level protocol on the supplied file. However, the same experiment shows 723 exact duplicates and 202 cross-split duplicate predictor cases. Random Forest and XGBoost both reach 1.0000 accuracy, while Logistic Regression and Naive Bayes do not. The model-specific contrast is consistent with the interpretation that flexible tree models can memorise repeated cases. The finding should therefore be interpreted as an evaluation-protocol result, not as evidence of clinical generalisation.

6.2 What Part 2 improves

The primary improvement is experimental validity. The proposed protocol removes exact duplicates before splitting, handles the documented out-of-range codes, represents nominal variables as categorical, keeps preprocessing and feature selection inside pipelines, retunes LASSE within every repeated training split and reports uncertainty over ten seeds. These choices directly address leakage and model-selection concerns described in the literature [5], [6]. The matched ten-seed comparison quantifies the effect: the paper-style protocol averages 99.71% accuracy versus 84.92% under the leakage-aware protocol, a 14.8-

point difference. The paper-style accuracy is higher in all 10 of 10 nominal seed cases; an unpaired Mann–Whitney U test gives $p < 0.001$, while the nominal Wilcoxon sensitivity calculation gives $p=0.0020$.

6.3 What Part 2 does not improve

LASSE does not produce a statistically significant accuracy improvement over the leakage-aware stacking baseline. On the primary holdout it ties the baseline on accuracy and slightly increases F1 and recall at the cost of specificity. The fair ablation also shows that the selector itself does not increase accuracy on that split. These negative findings are retained because they are important for a credible research report: the main contribution is a more defensible evaluation protocol, not an unsupported claim that the proposed classifier is better.

6.4 Strengths, limitations and practical implications

- Strength — The study separates faithful reproduction from methodological redesign, preserving the original protocol for Part 1 while preventing duplicate leakage in Part 2.
- Strength — Preprocessing, feature selection and tuning are performed inside training partitions, reducing information leakage and selection bias [5], [6].
- Strength — The fair ablation isolates the selector, while repeated evaluation and complementary statistical tests reduce reliance on a single split; the Nadeau–Bengio correction is used for the repeated comparison [8].
- Limitation — The 1,025 rows collapse to only 302 unique observations after deduplication, so the effective information content is much smaller than the nominal sample size.
- Limitation — The leakage-aware stacking baseline is not tuned to the same extent as LASSE. The comparison is therefore conservative and asymmetric.
- Limitation — The selector is stability-inspired rather than a full classical stability-selection implementation: five repetitions yield selection frequencies in increments of 0.2, whereas the original stability-selection literature uses substantially more resampling [7].
- Limitation — The public benchmark is not an independent clinical cohort. The results should not be interpreted as evidence of deployment-level diagnostic performance.
- Practical implication — If the objective is screening, LASSE’s higher recall on the primary clean holdout may be worth examining, but the associated reduction in specificity and lack of statistically significant accuracy improvement mean that further external validation is required before any clinical use.

7. Conclusion

This study reproduced the selected stacking-based heart-disease prediction paper and then evaluated its protocol under a duplicate-aware design. The seed-42 paper-style stacking model reaches 1.0000 accuracy, but the dataset contains 723 exact duplicate rows and 202 test observations with duplicate predictor vectors in training. The matched ten-seed analysis provides the stronger comparison: paper-style stacking averages 0.9971 ± 0.0062 accuracy, while leakage-aware stacking averages 0.8492 ± 0.0603 , a 14.8 percentage-point mean gap. The paper-style accuracy is higher in all 10 of 10 nominal seed cases, with an unpaired Mann–Whitney U $p < 0.001$; the Wilcoxon $p=0.0020$ is retained only as a

sensitivity calculation because the underlying test sets differ. The paper’s own confusion-matrix counts also reveal inconsistencies in its reported specificity and MCC.

LASSE provides a reproducible leakage-aware alternative combining duplicate control, domain-aware preprocessing, stability-inspired feature selection, tuned heterogeneous voting and repeated evaluation. It does not significantly outperform the leakage-aware stacking baseline, and the fair ablation shows no accuracy gain attributable to the selector. The final conclusion is therefore methodological: the apparent near-perfect performance is highly sensitive to the row-level evaluation protocol, and duplicate-aware evaluation gives a materially different estimate of generalisation.

8. Part 3 — Video presentation and reproducibility

The video presentation covers the selected paper, the faithful reproduction, the duplicate-leakage finding, the LASSE pipeline, executed code and outputs, comparative results, limitations and future work.

Video presentation: [Add final video link here]

Code/data archive: [Add final GitHub, OneDrive or Dropbox link here]

The package contains the clean notebook, executed notebook, supplied heart.csv, result CSV files, figures, requirements.txt and README.

9. GenAI acknowledgement

Generative AI tools were used during development for planning, debugging, structural editing and language refinement. The student reviewed and modified the resulting code and report, and the numerical results reported here were generated from the executed Python notebook using the supplied heart.csv file. The student remains responsible for understanding, verifying and submitting the final work in accordance with academic-integrity requirements.

References

- [1] M. Bhagat, A. Sharma, and P. Agarwal, “An efficient stacking-based ensemble technique for early heart attack prediction,” *Multimedia Tools and Applications*, vol. 84, pp. 36351–36375, 2025, doi: 10.1007/s11042-024-19293-7.
- [2] D. H. Wolpert, “Stacked generalization,” *Neural Networks*, vol. 5, no. 2, pp. 241–259, 1992, doi: 10.1016/S0893-6080(05)80023-1.
- [3] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2016, pp. 785–794, doi: 10.1145/2939672.2939785.
- [4] F. Pedregosa et al., “Scikit-learn: Machine learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [5] S. Kapoor and A. Narayanan, “Leakage and the reproducibility crisis in machine-learning-based science,” *Patterns*, vol. 4, no. 9, 100804, 2023, doi: 10.1016/j.patter.2023.100804.

- [6] G. C. Cawley and N. L. C. Talbot, “On over-fitting in model selection and subsequent selection bias in performance evaluation,” *Journal of Machine Learning Research*, vol. 11, pp. 2079–2107, 2010.
- [7] N. Meinshausen and P. Bühlmann, “Stability selection,” *Journal of the Royal Statistical Society: Series B*, vol. 72, no. 4, pp. 417–473, 2010, doi: 10.1111/j.1467-9868.2010.00740.x.
- [8] C. Nadeau and Y. Bengio, “Inference for the generalization error,” *Machine Learning*, vol. 52, no. 3, pp. 239–281, 2003, doi: 10.1023/A:1024068626366.

Appendix A — Reproducibility checklist

- Executed notebook included with visible outputs.
- Original supplied heart.csv included in the ZIP package.
- Part 1 retains the full dataset for faithful reproduction.
- Part 2 removes exact duplicates before the new split.
- Required metrics: Accuracy, Precision, Recall, F1 and AUC.
- Additional diagnostics: Specificity and MCC.
- Fair feature-selection ablation included.
- Primary tuning uses 30 sampled configurations per model family.
- Repeated evaluation retunes LASSE inside every training split.
- Matched ten-seed paper-style versus leakage-aware comparison included.
- Wilcoxon sensitivity calculation, Mann–Whitney U test and Nadeau–Bengio corrected resampled test included.
- Published-versus-reproduced table included.
- Paper Figure 5 metric consistency recomputed from its confusion-matrix counts.
- Transformed feature names shown in stability output.
- Feature set, classifier methods and metric definitions explicitly documented.
- Video presentation: [Add final video link here]
- Code/data archive: [Add final GitHub, OneDrive or Dropbox link here]